

A Practical Hierarchical Reinforcement Learning Framework for Slice-Aware Load Balancing with Directional A3 Offset Control and Bias-Interpreted Handover Admission

Version 15.0 – Directional Local Offset and Load-Aware Safe Admission Redesign

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Abstract

This document presents a redesigned hierarchical reinforcement learning (HRL) framework for slice-aware traffic coordination and mobility-safe A3 offset control in a multi-gNB O-RAN setting. The previous design used a global per-gNB per-slice bias and a local per-gNB per-slice A3 offset. That design is simple, but it creates two practical problems: first, the selected offset may be too weak to trigger handovers even when a neighbor is available; second, when the offset is made stronger, many UEs may hand over at once, causing total migration from a loaded cell to another cell rather than balanced load distribution.

The new design keeps the upper agent simple. The upper agent remains a global traffic coordinator and outputs only a per-gNB per-slice bias

$$b_{i,s}(t) \in [-1, 1],$$

where the sign indicates release or retain behavior, and the magnitude represents the intended traffic coordination pressure. The upper reward is a simple load-balance improvement reward based on the reduction of slice-load dispersion across gNBs.

The local level is redesigned. Each gNB still hosts exactly one local agent, but the local agent now outputs neighbor-specific and slice-specific directional A3 offsets

$$O_{i,j,s}(t), \quad j \in \mathcal{N}_i, s \in \mathcal{S}.$$

Thus, a gNB can encourage handover toward a lightly loaded neighbor while discouraging handover toward a congested neighbor.

Finally, a deterministic safe admission layer is inserted after A3 eligibility and before handover execution. This layer interprets the absolute negative value of the upper bias, $\max(0, -b_{i,s})$, as an offloading pressure and converts it into an admission capacity. The A3 offset determines which UEs are radio-eligible, while the admission layer determines how many eligible UEs may actually be admitted to each target neighbor without overloading the target slice. This separation solves both problems: stronger offsets can be used when necessary, while admission control prevents aggressive total migration.

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1 High-Level Idea

The objective is to balance slice traffic across multiple gNBs while preserving mobility robustness. The redesigned framework separates the decision into three layers:

1. **Global traffic coordination:** decide which gNB-slice pairs should release or retain traffic, and how strong this coordination pressure should be.
2. **Local directional mobility execution:** for each serving gNB, decide directional A3 offsets toward each neighbor and slice.
3. **Safe handover admission:** after A3 eligibility, admit only the best feasible handovers that keep the target slice load safe.

The architecture is:

Global PPO coordinator $\Rightarrow b_{i,s} \Rightarrow$ Local gNB agent $\Rightarrow O_{i,j,s} \Rightarrow$ A3 eligibility $\Rightarrow$ Safe admission $\Rightarrow$ HO execution

The key design change is that the A3 offset no longer controls both eligibility and migration volume. Instead:

Directional A3 offset controls radio eligibility.

Safe admission controls accepted handover intensity.

2 Motivation for the Redesign

The previous design used one local offset per serving gNB and slice:

$$O_{i,s}(t).$$

This means that if gNB i makes handovers easier for slice s , the same offset is applied toward all neighbors. This creates two problems.

2.1 Problem 1: Weak Offset

The A3 rule is:

$$RSRP_{u,j}(t) > RSRP_{u,i}(t) + O_{i,s}(t) + H_{A3}.$$

If the strongest available negative offset is only -6 dB, the UE may still not hand over when the target is in coverage but several dB weaker than the serving gNB. For example, if $H_{A3} = 1$ dB and $O_{i,s} = -6$ dB, then the condition becomes:

$$RSRP_{u,j}(t) > RSRP_{u,i}(t) - 5 \text{ dB}.$$

If the target is 8 dB weaker, the UE remains attached to the source even when the source is congested.

2.2 Problem 2: Total Migration

If stronger offsets are used without admission control, many UEs can become A3-eligible at the same time. This may empty the overloaded source cell and overload the target cell. In that case, the controller does not really balance load; it transfers congestion from one cell to another.

Therefore, the new design explicitly separates:

- **who can move** – determined by the directional A3 offset and radio measurements;
- **how many can move** – determined by the safe admission layer using the bias magnitude and target headroom.

3 System Model

Let:

$$\mathcal{G} = \{1, 2, \dots, N\}$$

be the set of gNBs, and:

$$\mathcal{S} = \{\text{eMBB}, \text{URLLC}, \text{mMTC}\}$$

be the set of network slices.

For each serving gNB $i \in \mathcal{G}$, let:

$$\mathcal{N}_i \subset \mathcal{G} \setminus \{i\}$$

be its candidate neighbor set.

3.1 Slice Load

For every gNB i and slice s , the normalized PRB load is:

$$L_{i,s}(t) = \frac{PRB_{i,s}^{used}(t)}{PRB_{i,s}^{total}(t)} \in [0, 1].$$

3.2 Connected UE Counter

The number of connected UEs of slice s at gNB i is:

$$K_{i,s}(t) = \sum_{u \in \mathcal{U}_i(t)} \mathbf{1}[\text{slice}(u) = s].$$

A normalized version may be used in learning:

$$K_{i,s}^{norm}(t) = \frac{K_{i,s}(t)}{K_{max}}.$$

3.3 Slice Average Load

For each slice, define the average load across the network:

$$\bar{L}_s(t) = \frac{1}{|\mathcal{G}|} \sum_{i \in \mathcal{G}} L_{i,s}(t).$$

This value acts as a simple dynamic target. A gNB-slice above $\bar{L}_s$ should tend to release traffic, while a gNB-slice below $\bar{L}_s$ may safely receive traffic if radio and mobility conditions allow.

3.4 Mobility Counters

For each source-target-slice direction, the simulator may track:

- $N_{i,j,s}^{HOA}$: handover attempts from gNB i to gNB j for slice s ;
- $N_{i,j,s}^{HOS}$: successful handovers;
- $N_{i,j,s}^{HOF}$: handover failures;
- $N_{i,j,s}^{PP}$: ping-pong handovers.

The ratios are:

$$R_{i,j,s}^{HF}(t) = \frac{N_{i,j,s}^{HOF}(t)}{N_{i,j,s}^{HOS}(t) + \epsilon},$$

$$R_{i,j,s}^{PP}(t) = \frac{N_{i,j,s}^{PP}(t)}{N_{i,j,s}^{HOS}(t) + \epsilon}.$$

Aggregated per-gNB versions may be used in a simpler implementation:

$$R_i^{HF}(t), \quad R_i^{PP}(t).$$

4 Directional A3 Handover Model

The local agent for serving gNB i outputs a neighbor-specific and slice-specific offset:

$$O_{i,j,s}(t), \quad j \in \mathcal{N}_i, \quad s \in \mathcal{S}.$$

A UE u served by gNB i , belonging to slice s , is A3-eligible toward neighbor j if:

$$RSRP_{u,j}(t) > RSRP_{u,i}(t) + O_{i,j,s}(t) + H_{A3}.$$

The condition must be held for TTT consecutive steps:

$$\sum_{\tau=t-TTT+1}^t \mathbf{1}[RSRP_{u,j}(\tau) > RSRP_{u,i}(\tau) + O_{i,j,s}(\tau) + H_{A3}] = TTT.$$

The A3 margin is:

$$m_{u,i,j,s}(t) = RSRP_{u,j}(t) - RSRP_{u,i}(t) - O_{i,j,s}(t) - H_{A3}.$$

If several UEs are eligible, the safe admission layer should prioritize the largest margin values.

4.1 Offset Semantics

Negative offsets make handover easier, and positive offsets make handover harder:

Offset	Effect	Meaning
$O_{i,j,s} < 0$	easier A3 condition	encourage handover from i to j for slice s
$O_{i,j,s} = 0$	neutral	default A3 behavior
$O_{i,j,s} > 0$	harder A3 condition	discourage handover from i to j for slice s

5 Upper Agent: Simple Global Bias Coordinator

5.1 Role

The upper agent is intentionally kept simple. It answers:

For each gNB and slice, should this gNB-slice release or retain traffic, and how strong should the pressure be?

It does not choose target neighbors and does not execute handovers.

5.2 Upper State

A minimal practical state is:

$$s_H(t) = \bigoplus_{i \in \mathcal{G}} \bigoplus_{s \in \mathcal{S}} \left[L_{i,s}(t), K_{i,s}^{norm}(t), b_{i,s}^{prev}(t) \right].$$

This gives:

$$\dim(s_H) = 3NS.$$

For $N = 3$ gNBs and $S = 3$ slices:

$$\dim(s_H) = 27.$$

An extended version may add SLA flags or mobility feasibility, but the recommended first implementation keeps the upper agent simple.

5.3 Upper Action

The upper action is the bias matrix:

$$a_H(t) = B(t) = [b_{i,s}(t)]_{i \in \mathcal{G}, s \in \mathcal{S}}.$$

Each entry satisfies:

$$b_{i,s}(t) \in [-1, 1].$$

The interpretation is:

Bias value	Meaning	Expected behavior
$b_{i,s} < 0$	release/offload	local agent should make outgoing handovers easier
$b_{i,s} \approx 0$	neutral	local agent should use near-neutral offset
$b_{i,s} > 0$	retain/protect	local agent should make outgoing handovers harder

The magnitude $|b_{i,s}|$ represents coordination pressure. In particular, the negative part:

$$P_{i,s}^{off}(t) = \max(0, -b_{i,s}(t))$$

is used by the safe admission layer as offloading pressure.

6 Simple Upper Reward

The upper agent should learn to reduce load imbalance across gNBs for each slice. Define the global load-balance cost:

$$J_H(t) = \sum_{s \in \mathcal{S}} \sum_{i \in \mathcal{G}} (L_{i,s}(t) - \bar{L}_s(t))^2.$$

This is proportional to the sum of per-slice load variances. If the loads are balanced, this cost is small.

The simplest upper reward is:

$$r_H(t) = J_H(t) - J_H(t + T_H)$$

where T_H is the upper control window. This reward is positive when the global slice-load imbalance decreases.

An optional smoothing penalty can be added after the simple version works:

$$r_H(t) = J_H(t) - J_H(t + T_H) - \lambda_\Delta \sum_{i \in \mathcal{G}} \sum_{s \in \mathcal{S}} (b_{i,s}(t) - b_{i,s}(t - T_H))^2$$

A practical initial value is:

$$\lambda_\Delta \in [0.01, 0.05].$$

7 Local Agent: One Agent per gNB with Directional Offsets

7.1 Role

Each gNB i hosts one local agent. This agent controls outgoing mobility from gNB i . It does not control incoming mobility and does not control other gNBs.

The local agent answers:

Given the upper bias and local neighbor conditions, which directional A3 offset should be applied toward each neighbor and slice?

7.2 Local State

For every neighbor $j \in \mathcal{N}_i$ and slice $s \in \mathcal{S}$, a practical local observation block is:

$$o_{i,j,s}(t) = [b_{i,s}(t), b_{j,s}(t), L_{i,s}(t), L_{j,s}(t), H_{j,s}(t), M_{i,j,s}(t), \Delta RSRP_{i,j,s}(t), O_{i,j,s}^{prev}(t), R_{i,j,s}^{HF}(t), R_{i,j,s}^{PP}(t)].$$

Here:

$$H_{j,s}(t) = \max(0, \bar{L}_s(t) - L_{j,s}(t))$$

is the target headroom relative to the slice average, and $M_{i,j,s}$ is the directional mobility feasibility fraction.

The full local state is:

$$o_i(t) = \bigoplus_{j \in \mathcal{N}_i} \bigoplus_{s \in \mathcal{S}} o_{i,j,s}(t).$$

7.3 Local Action

The local action is:

$$a_{L,i}(t) = [O_{i,j,s}(t)]_{j \in \mathcal{N}_i, s \in \mathcal{S}}.$$

The action dimension is:

$$\dim(a_{L,i}) = |\mathcal{N}_i| |\mathcal{S}|.$$

For three gNBs and three slices, each gNB has two neighbors, so:

$$\dim(a_{L,i}) = 2 \times 3 = 6.$$

The offset set may be:

$$\mathcal{A}_{offset} = \{-12, -10, -8, -6, -4, -2, 0, +2, +4, +6\} \text{ dB}.$$

For early debugging, a smaller set is recommended:

$$\{-6, -4, -2, 0, +2, +4, +6\} \text{ dB}.$$

The extended negative values $-8, -10, -12$ should be enabled only after the safe admission layer is active.

8 Local Reward

The local reward should encourage safe execution of the upper bias. A simple local reward is:

$$r_{L,i}(t) = w_{align} R_i^{align}(t) + w_{bal} R_i^{bal}(t) - w_{over} P_i^{over}(t) - w_{HO} P_i^{HO}(t) - w_{HF} R_i^{HF}(t) - w_{PP} R_i^{PP}(t) - w_{\Delta O} P_i^{\Delta O}(t).$$

8.1 Bias-Offset Alignment

Normalize the offset by 12:

$$\bar{O}_{i,j,s}(t) = \frac{O_{i,j,s}(t)}{12}.$$

The alignment reward is:

$$R_i^{align}(t) = \frac{1}{|\mathcal{N}_i| |\mathcal{S}|} \sum_{j \in \mathcal{N}_i} \sum_{s \in \mathcal{S}} b_{i,s}(t) \bar{O}_{i,j,s}(t).$$

If $b_{i,s} < 0$ and $O_{i,j,s} < 0$, the product is positive. If $b_{i,s} > 0$ and $O_{i,j,s} > 0$, the product is also positive.

8.2 Local Neighborhood Balance Improvement

Define the local neighborhood:

$$\mathcal{C}_i = \{i\} \cup \mathcal{N}_i.$$

The local slice-load variance is:

$$Var_{\mathcal{C}_i,s}(t) = \frac{1}{|\mathcal{C}_i|} \sum_{k \in \mathcal{C}_i} (L_{k,s}(t) - \bar{L}_{\mathcal{C}_i,s}(t))^2,$$

where:

$$\bar{L}_{\mathcal{C}_i,s}(t) = \frac{1}{|\mathcal{C}_i|} \sum_{k \in \mathcal{C}_i} L_{k,s}(t).$$

Then:

$$R_i^{bal}(t) = \sum_{s \in \mathcal{S}} [Var_{\mathcal{C}_i,s}(t) - Var_{\mathcal{C}_i,s}(t + T_L)].$$

8.3 Target Overload Penalty

$$P_i^{over}(t) = \sum_{j \in \mathcal{N}_i} \sum_{s \in \mathcal{S}} \left[\max \left(0, L_{j,s}(t + T_L) - L_s^{safe} \right) \right]^2.$$

8.4 Handover Cost and Mobility Penalties

Let $N_i^{HO}(t)$ be the number of handovers leaving gNB i during the local window, and let $K_i(t) = \sum_s K_{i,s}(t)$. Then:

$$P_i^{HO}(t) = \frac{N_i^{HO}(t)}{K_i(t) + \epsilon}.$$

The offset smoothness penalty is:

$$P_i^{\Delta O}(t) = \sum_{j \in \mathcal{N}_i} \sum_{s \in \mathcal{S}} (\bar{O}_{i,j,s}(t) - \bar{O}_{i,j,s}(t - T_L))^2.$$

A practical starting reward is:

$$r_{L,i}(t) = 0.5R_i^{align}(t) + 2.0R_i^{bal}(t) - 3.0P_i^{over}(t) - 0.2P_i^{HO}(t) - 2.0R_i^{HF}(t) - 3.0R_i^{PP}(t) - 0.05P_i^{\Delta O}(t)$$

For early training, start with:

$$r_{L,i}^{(1)}(t) = 0.5R_i^{align}(t),$$

then add neighborhood balance, then add safety penalties.

9 Safe Admission Layer

9.1 Purpose

The safe admission layer is a deterministic filter inserted after A3 eligibility and before handover execution. It is not a learned policy. It protects the system from total migration and target-cell overload.

The final handover pipeline is:

$$O_{i,j,s} \Rightarrow \text{A3 eligibility} \Rightarrow \text{safe admission} \Rightarrow \text{HO execution}.$$

9.2 Offloading Pressure from Bias

The upper bias is interpreted as:

$$P_{i,s}^{off}(t) = \max(0, -b_{i,s}(t)).$$

If $b_{i,s} = -0.2$, then $P_{i,s}^{off} = 0.2$, meaning weak offload pressure. If $b_{i,s} = -0.9$, then $P_{i,s}^{off} = 0.9$, meaning strong offload pressure. If $b_{i,s} > 0$, then $P_{i,s}^{off} = 0$, meaning no offload admission from that source slice.

9.3 Source Excess and Target Headroom

Using the slice average load $\bar{L}_s(t)$, define source excess:

$$E_{i,s}(t) = \max(0, L_{i,s}(t) - \bar{L}_s(t)).$$

Define target headroom:

$$H_{j,s}(t) = \max(0, \bar{L}_s(t) - L_{j,s}(t)).$$

Only overloaded source slices can send traffic, and only below-average target slices can receive traffic.

9.4 Admission Capacity

The maximum accepted handovers from source i to target j for slice s in the local window is:

$$A_{i,j,s}(t) = \left\lceil P_{i,s}^{off}(t) \cdot \min(E_{i,s}(t), H_{j,s}(t)) \cdot K_{i,s}(t) \right\rceil$$

This formula converts the upper bias magnitude into an admission intensity. It also prevents over-admission by considering both source excess and target headroom.

9.5 Hard Target Safety

Let L_s^{safe} be a hard load limit, for example 0.75 or 0.80. For a candidate UE u , estimate the target load after handover:

$$L_{j,s}^{after}(t) = L_{j,s}(t) + \Delta L_{u,j,s}.$$

A simple approximation is:

$$\Delta L_{u,j,s} \approx \frac{PRB_u}{PRB_{j,s}^{total}}.$$

The handover is accepted only if:

$$L_{j,s}^{after}(t) \leq L_s^{safe}.$$

9.6 Final Handover Acceptance Rule

A handover $u : i \rightarrow j$ for slice s is executed only if all conditions hold:

$$RSRP_{u,j}(t) > RSRP_{u,i}(t) + O_{i,j,s}(t) + H_{A3}$$

held for TTT consecutive steps,

$$P_{i,s}^{off}(t) > 0, \quad E_{i,s}(t) > 0, \quad H_{j,s}(t) > 0,$$

$$N_{i,j,s}^{accepted}(t) < A_{i,j,s}(t),$$

and:

$$L_{j,s}^{after}(t) \leq L_s^{safe}.$$

Among all A3-eligible candidates, UEs should be sorted by the A3 margin $m_{u,i,j,s}$, and the best radio candidates should be admitted first.

9.7 Pseudocode

```
accepted_counts = {}

for source_id in gnbs:
    for target_id in neighbors[source_id]:
        for slice_type in slices:
            candidates = []

            for ue in ues_served_by(source_id, slice_type):
                if not a3_condition_satisfied(ue, source_id, target_id, slice_type):
                    continue
                margin = compute_a3_margin(ue, source_id, target_id, slice_type)
                candidates.append((ue, margin))

            candidates.sort(key=lambda item: item[1], reverse=True)

            for ue, margin in candidates:
                if safe_admission(
                    ue=ue,
                    source_id=source_id,
                    target_id=target_id,
                    slice_type=slice_type,
                    bias=B,
                    loads=L,
                    ue_counts=K,
                    accepted_counts=accepted_counts,
                ):
                    perform_handover(ue, source_id, target_id)
```

10 Step-by-Step Example

Consider three gNBs and one slice, eMBB:

gNB	eMBB load
gNB0	0.90
gNB1	0.40
gNB2	0.50

The slice average is:

$$\bar{L}_{eMBB} = \frac{0.90 + 0.40 + 0.50}{3} = 0.60.$$

The upper agent outputs:

$$b_{0,eMBB} = -0.8.$$

Thus:

$$P_{0,eMBB}^{off} = 0.8.$$

For the direction gNB0 $\rightarrow$ gNB1:

$$E_{0,eMBB} = 0.90 - 0.60 = 0.30,$$

$$H_{1,eMBB} = 0.60 - 0.40 = 0.20.$$

If $K_{0,eMBB} = 10$, the admission capacity is:

$$A_{0,1,eMBB} = \lceil 0.8 \times \min(0.30, 0.20) \times 10 \rceil = \lceil 1.6 \rceil = 2.$$

Therefore, even if many UEs are A3-eligible, only two eMBB handovers from gNB0 to gNB1 are admitted in the current local control window.

11 Training Workflow

The recommended training workflow is:

1. **Simulator sanity checks.** Verify that negative offsets increase A3 eligibility and positive offsets reduce it. Verify that the safe layer limits accepted handovers.
2. **Local pretraining with rule-based biases.** Use a simple rule for $b_{i,s}$ and train local agents to map bias plus neighbor information to directional offsets.
3. **Upper training with frozen local agents.** Freeze local policies and train the upper PPO using the simple global load-balance reward.
4. **Joint fine-tuning.** Fine-tune both levels with smaller learning rates.

A simple rule-based bias for pretraining is:

$$b_{i,s}^{rule}(t) = \begin{cases} -1, & L_{i,s}(t) > \bar{L}_s(t) + \delta, \\ +1, & L_{i,s}(t) < \bar{L}_s(t) - \delta, \\ 0, & \text{otherwise.} \end{cases}$$

where δ is a small deadband, for example 0.05.

12 Validation

12.1 Simulator Sanity Checks

Before learning, the simulator should satisfy:

- negative directional offsets increase A3 eligibility toward the corresponding target neighbor;
- positive directional offsets reduce A3 eligibility toward the corresponding target neighbor;
- the A3 condition includes offset, hysteresis, and TTT;
- the same source gNB can encourage handover toward one neighbor and discourage handover toward another;
- the safe layer limits accepted handovers according to $A_{i,j,s}$;
- target load does not exceed L_s^{safe} after admitted handovers;
- successful handovers update serving gNB, UE counts, and slice loads consistently.

12.2 Upper-Agent Validation

The upper agent is valid if it reduces:

- global slice-load dispersion J_H ;
- overloaded gNB-slice pairs;
- unnecessary strong bias oscillations.

12.3 Local-Agent Validation

The local agent is valid if:

- for $b_{i,s} < 0$, it tends to choose negative offsets toward safe neighbors;
- for $b_{i,s} > 0$, it tends to choose neutral or positive offsets;
- it avoids sending traffic to overloaded targets;
- handover failure and ping-pong ratios remain controlled.

13 Baselines

The final evaluation should compare:

1. fixed default A3 offset;
2. neutral global bias $b_{i,s} = 0$;
3. rule-based bias with rule-based directional offsets;
4. learned upper bias with rule-based directional offsets;
5. learned upper bias with learned directional local offsets, without safe admission;
6. full proposed design: learned upper bias, learned directional local offsets, and safe admission.

The last comparison is important because it shows whether the admission layer actually prevents total migration.

14 Complexity Analysis

14.1 Upper Agent

The upper action dimension remains:

$$\dim(a_H) = NS.$$

For three gNBs and three slices:

$$\dim(a_H) = 9.$$

Thus, the upper agent stays simple.

14.2 Local Agents

Each local agent has action dimension:

$$\dim(a_{L,i}) = |\mathcal{N}_i|S.$$

With three gNBs, each gNB has two neighbors and three slices:

$$\dim(a_{L,i}) = 6.$$

If the offset set has 10 discrete values, a factorized categorical policy outputs:

$$10|\mathcal{N}_i|S$$

logits. For two neighbors and three slices:

$$10 \times 2 \times 3 = 60$$

logits per local agent. This avoids a flat joint action with 10^6 classes.

15 O-RAN Integration

O-RAN component	Role
Non-RT RIC rApp	Hosts upper PPO and computes $b_{i,s}$ every T_H
A1 interface	Sends coordination bias policies to near-RT RIC
near-RT RIC xApp	Hosts one local agent per gNB
E2SM-KPM	Reports load, UE counts, and mobility KPIs
E2SM-RC	Sends directional slice-specific A3 offset configurations if supported
O-CU-CP	Applies A3 offset configuration and executes handover control
O-DU	Provides radio measurements such as RSRP

15.1 Example Control Message

```
{
  "action_type": "DIRECTIONAL_SLICE_A3_OFFSET_CONFIG",
  "source_gNB": "gNB-0",
  "slice_configs": [
    {
      "slice": "eMBB",
      "global_bias": -0.80,
      "offload_pressure": 0.80,
```

```

    "neighbors": [
      {
        "target_gNB": "gNB-1",
        "applied_offset_dB": -8,
        "admission_capacity": 2
      },
      {
        "target_gNB": "gNB-2",
        "applied_offset_dB": -4,
        "admission_capacity": 1
      }
    ]
  },
  {
    "slice": "URLLC",
    "global_bias": 0.30,
    "offload_pressure": 0.00,
    "neighbors": [
      {
        "target_gNB": "gNB-1",
        "applied_offset_dB": 2,
        "admission_capacity": 0
      },
      {
        "target_gNB": "gNB-2",
        "applied_offset_dB": 4,
        "admission_capacity": 0
      }
    ]
  }
],
"a3_hysteresis_dB": 1.0,
"ttt_steps": 3,
"validity_duration_ms": 500
}

```

16 Implementation Notes

- Keep the upper agent simple: state $[L, K^{norm}, b^{prev}]$, action $b_{i,s}$, reward $J_H(t) - J_H(t+T_H)$.
- Use one local agent per gNB, not one local agent per link.
- Change local offsets from $O_{i,s}$ to $O_{i,j,s}$.
- Enable strong offsets such as $-8, -10, -12$ only after the safe admission layer is active.
- The safe layer should be deterministic, not learned.
- Interpret $\max(0, -b_{i,s})$ as offloading pressure.
- Use source excess and target headroom relative to the slice average load.
- Sort A3-eligible UEs by A3 margin and admit the best radio candidates first.
- Evaluate with the same traffic and mobility seeds against a neutral-bias baseline.

17 Conclusion

This Version 15 redesign keeps the hierarchy meaningful while solving the two practical problems observed in simulation. The upper agent remains simple and useful: it learns a global per-gNB

per-slice bias that represents release or retain pressure. The local agent remains decentralized with one agent per gNB, but it becomes directional by outputting neighbor-specific slice-specific offsets $O_{i,j,s}$. The deterministic safe admission layer converts the negative bias magnitude into an admitted handover intensity and prevents total migration by checking source excess, target headroom, per-direction admission capacity, and hard target-load safety.

The resulting framework can be summarized as:

Upper bias $b_{i,s}$: global load pressure

Local offset $O_{i,j,s}$: directional A3 eligibility

Safe admission $A_{i,j,s}$: controlled handover intensity

This structure preserves the original HRL concept while making the bias magnitude operationally meaningful and preventing uncontrolled offloading.

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